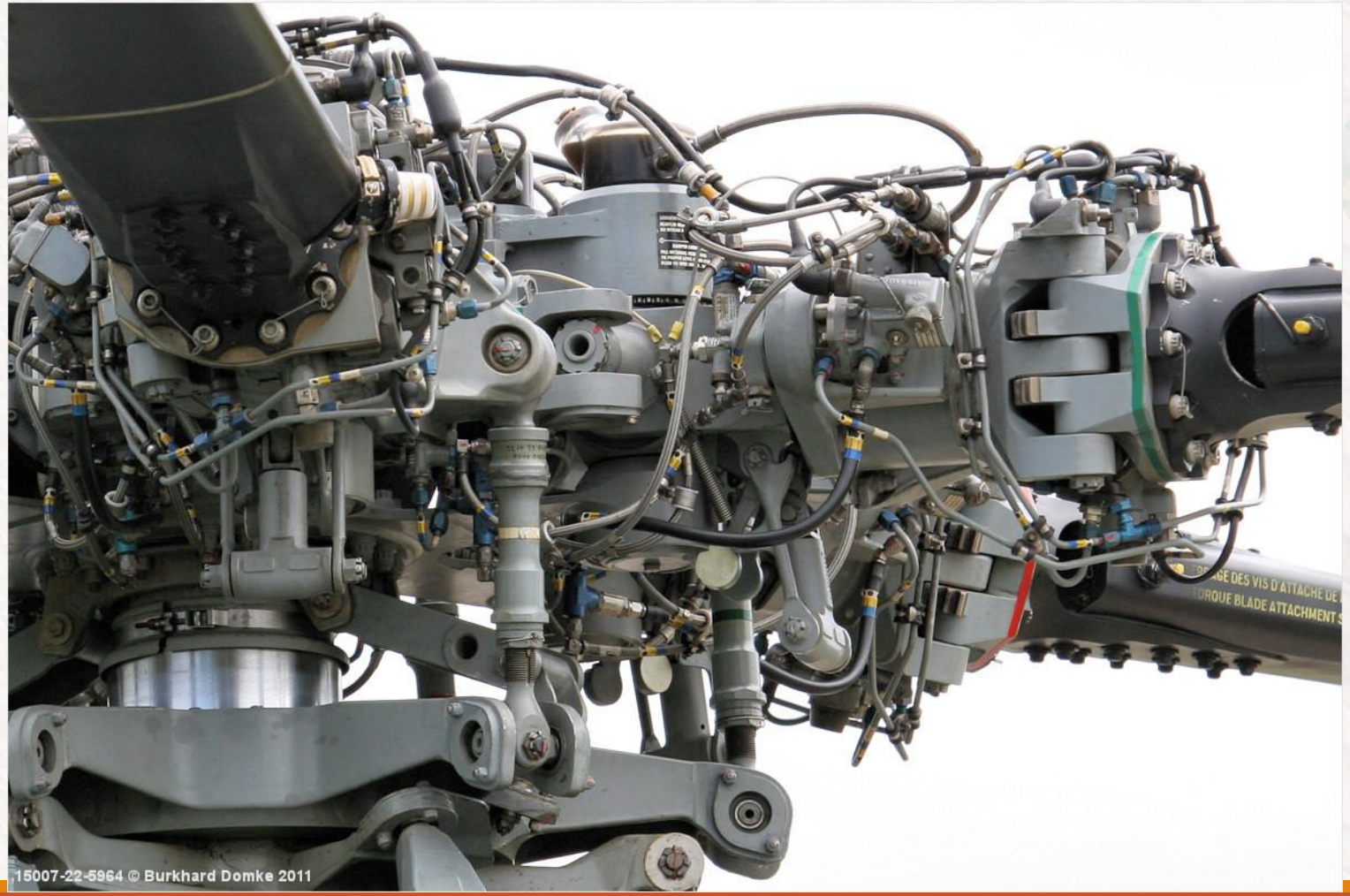
Rotary Wing Aerodynamics (AE 667)

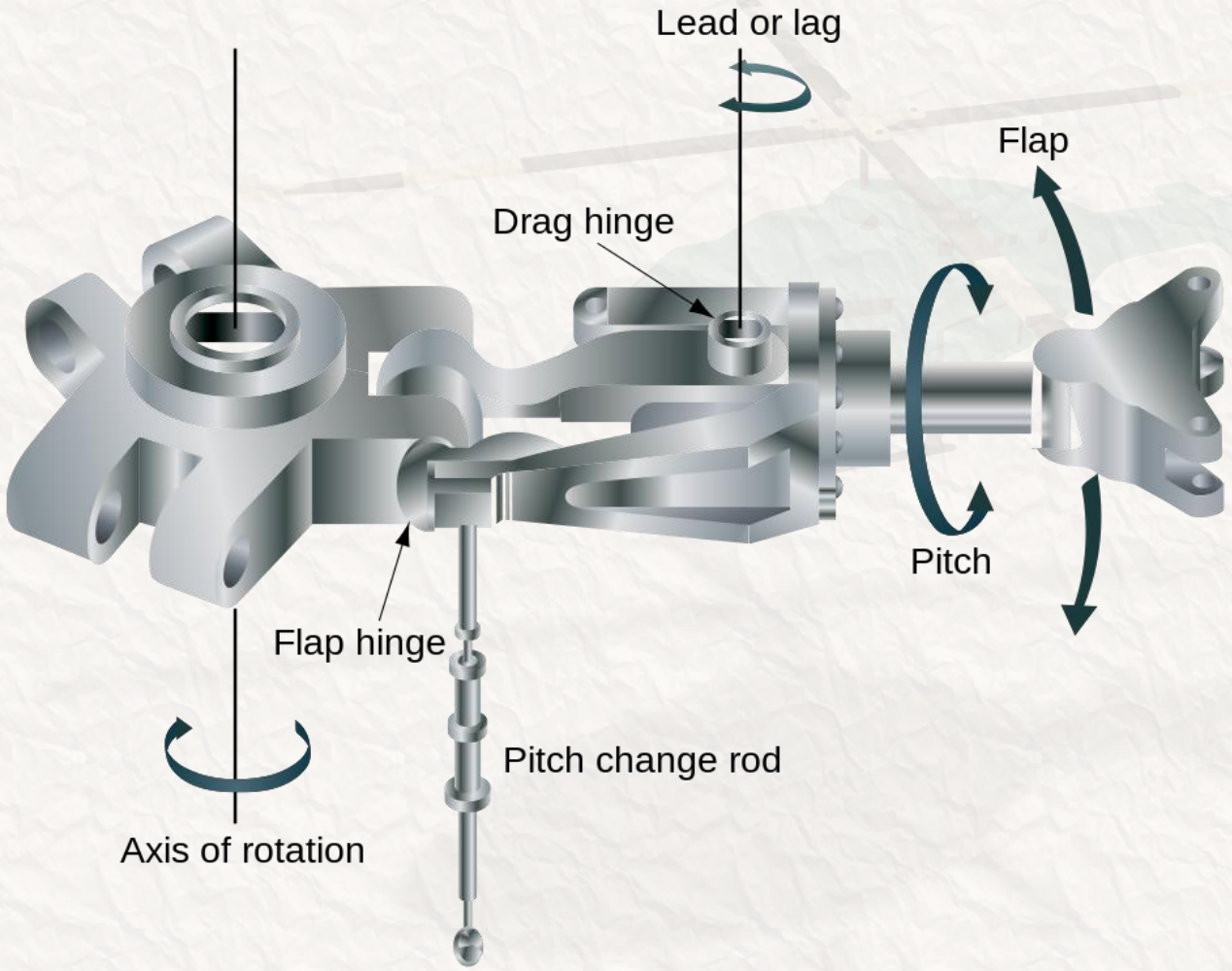


Blade Dynamics

Helicopter Rotor Hub



Helicopter Rotor Hub

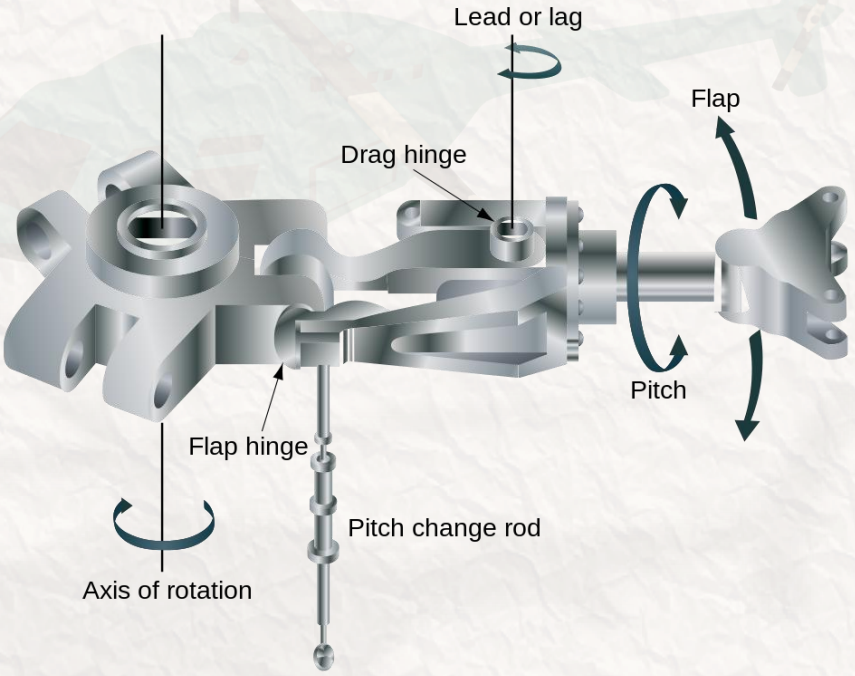


Helicopter Rotor Hub degrees of freedom

- Flapping hinges: Allows blade to flap up or down. Avoids transmission of bending moments to the shaft

- Lead-Lag hinges: Avoids stresses at the blade root due to Coriolis forces.
  (Lead-lag motion neglected from aerodynamic perspective)

- Pitch / Feathering (bearing, swash plate, pitch links): Control blade pitch for controlling helicopter motion.



Hub type: Rigid

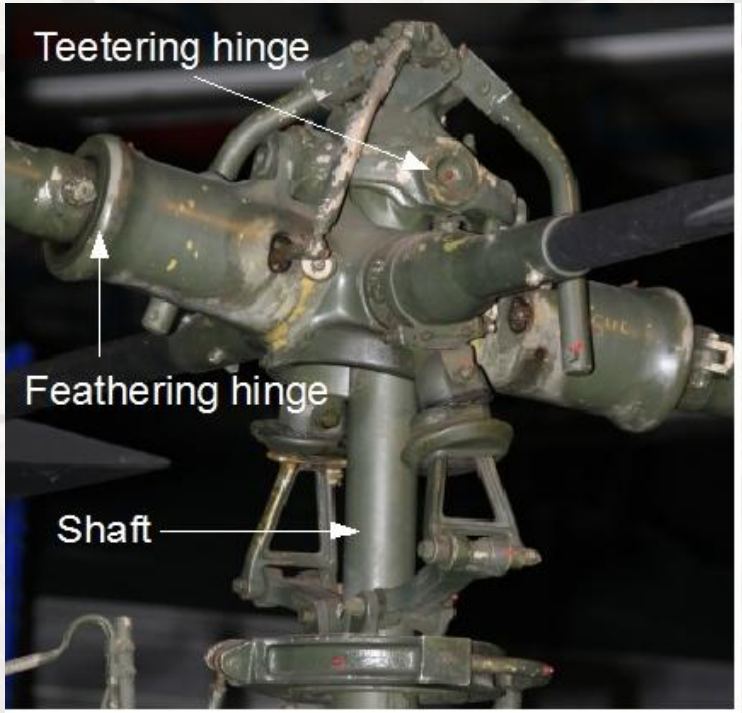
- Only feathering (pitching), No Flapping
- Moments transmitted to the rotor head
- High degree of control power.
- Fewer parts ➔ less maintenance



http://www.helistart.com/RotorheadTypes.aspx

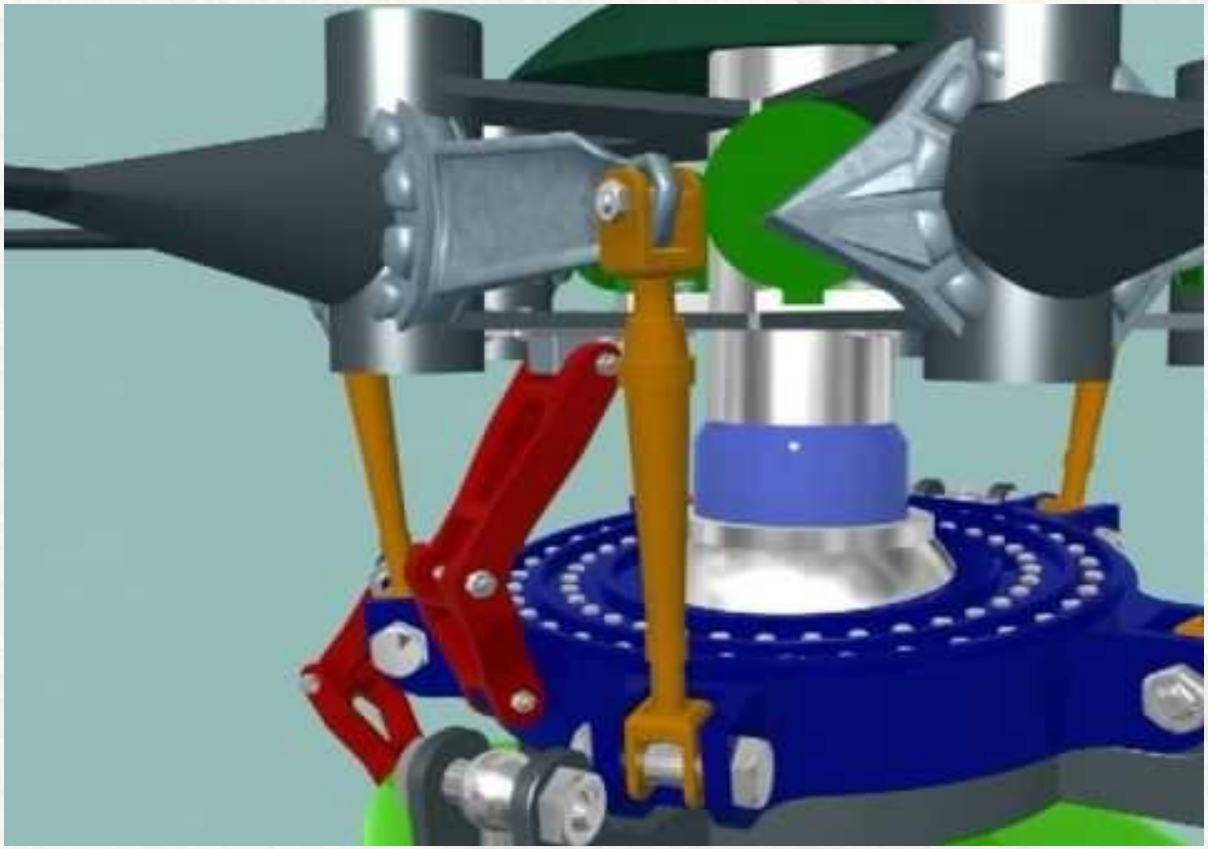
Hub type: Teetering

- Blades connected together through a common pivot
- Blades fixed at an average coning angle to reduce stress
- Rotor disc can tilt in any direction
- Lesser degree of control compared to rigid system



http://www.helistart.com/RotorheadTypes.aspx

Swash Plate Animation

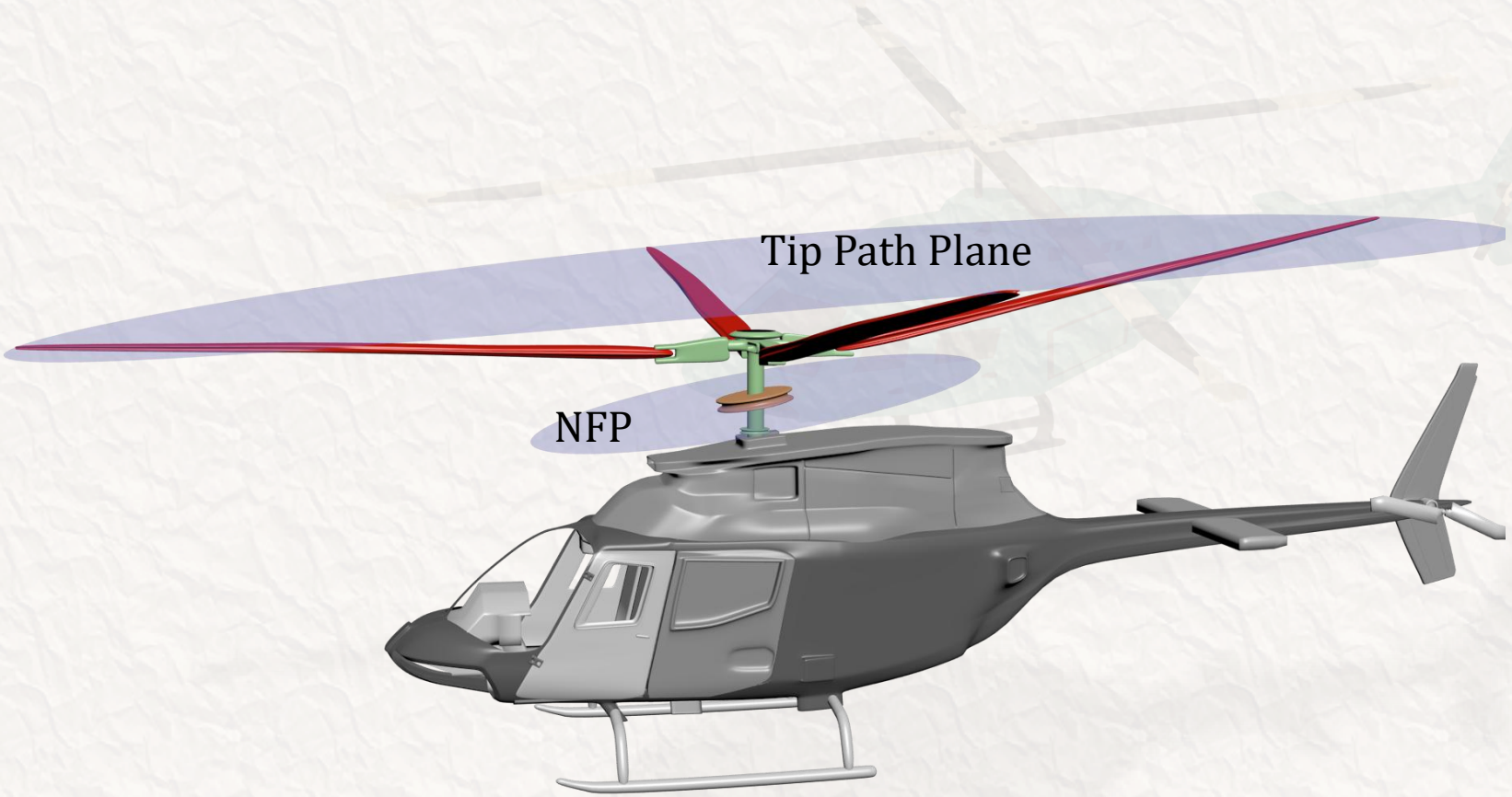


https://www.youtube.com/watch?v=83h6QK-oJ4M

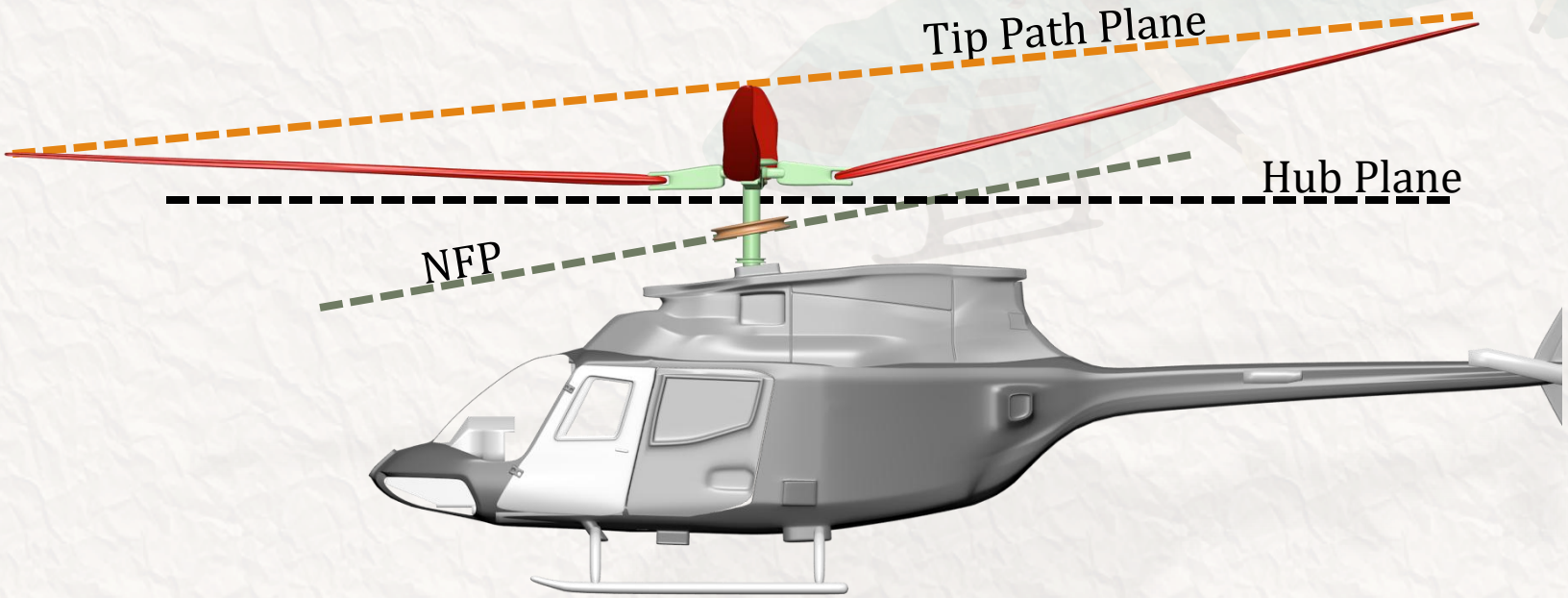
Coordinate Systems

- Hub Plane (Shaft Plane)
  - 'Z' axis along rotor shaft
  - Rotor is seen flapping and feathering (changes pitch) about this plane
- Tip Path Plane (TPP)
  - Plane in which the tips of the rotor blades move
  - Rotor is seen to be only feathering. No flapping motion
- No Feathering Plane (NFP)
  - Pitch constant when seen from this plane
  - Rotor is seen to be flapping only

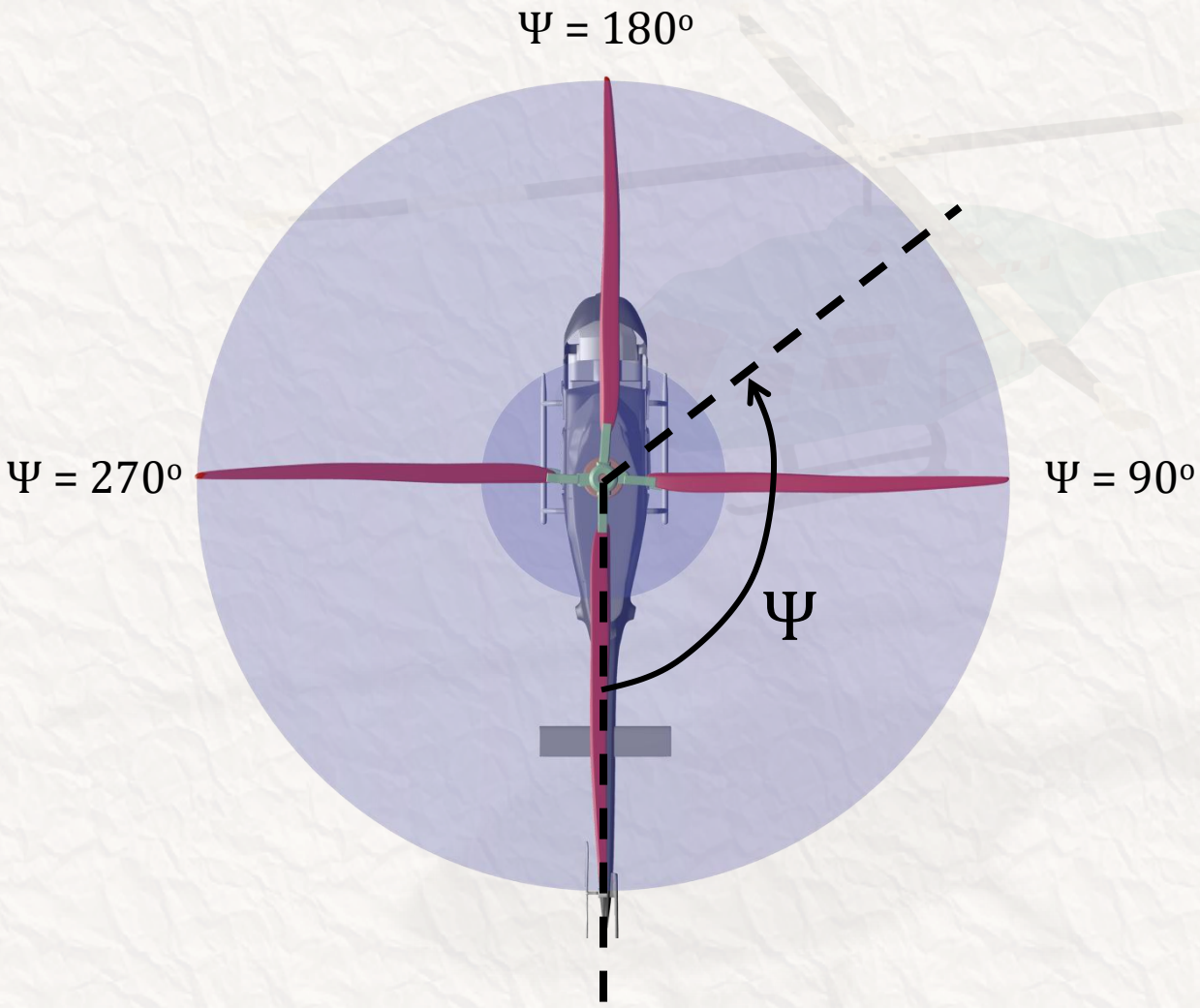
Coordinate Systems



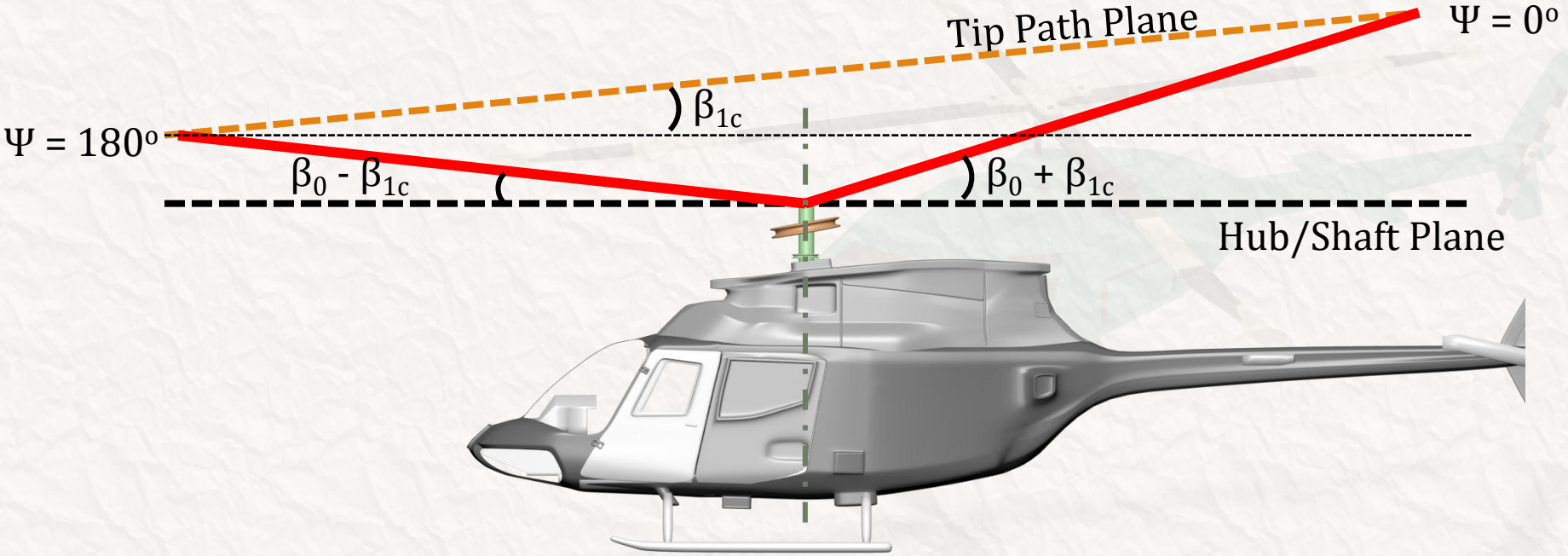
Coordinate Systems



Azimuthal angles



Tip Path Plane &Flapping Angle



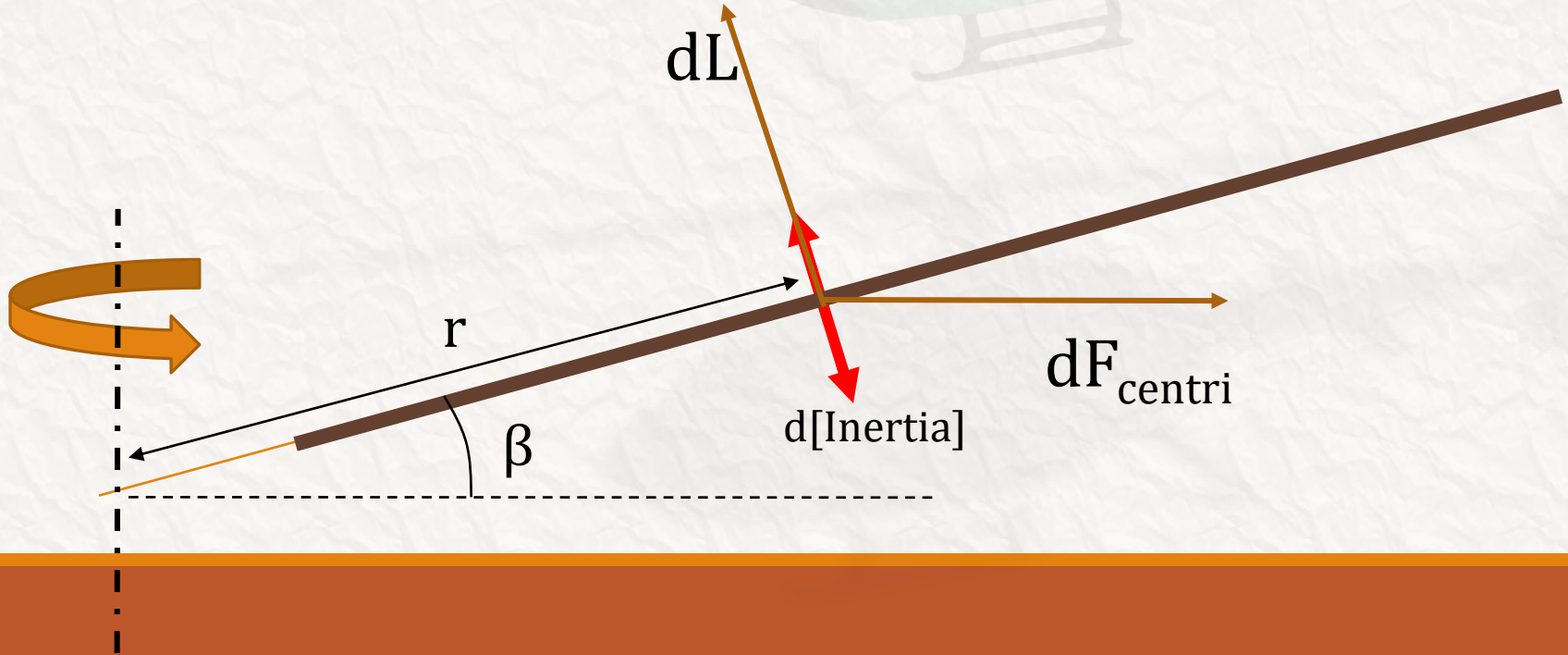
$$\beta\,(\text{Flapping Angle}) \cong \beta_0 + \beta_{1c}\cos(\psi) + \beta_{1s}\sin(\psi)$$

Higher order terms/harmonics relevant while studying vibration

Blade Dynamics
(Blade Flapping Dynamics)

Forces and Moments on Rotor Blade

- We assume 0 hinge offset for simplicity
- Steady forward flight: Net moments at hinge due to aerodynamic, centrifugal, & inertial forces cancel out to achieve equilibrium



Moment Balance

- Counter-clockwise moments
  - Aerodynamic:

$$\int_{r=0}^{r=R} \frac{1}{2} \rho c \left(\Omega r + V_\infty \sin\psi\right)^2 r C_l dr$$

(as derived while studying hover coning)

- Clockwise moments:
  - Centrifugal: $I\Omega^2\beta$ (as derived while studying hover coning)
  - Inertial
    (due to angular acceleration in flapping direction): $I\ddot{\beta}$

Moment Balance

- Clockwise Moments = Counter-clockwise moments

$$I\ddot{\beta} + I\Omega^2\beta = \int_{r=0}^{r=R} \frac{1}{2}\rho c\left(\Omega r + V_\infty \sin\psi\right)^2 r C_l dr$$

- The equation resembles a harmonic oscillator with natural frequency of Ω and the forcing term (right hand side) also consisting of first, second, and higher harmonics of Ω

- Aerodynamic damping ensures stability of the otherwise resonant system

Solving ODE (Partially)

- Assume forcing function (RHS) of the form: Asin(Ωt)
- Assume a damping term in LHS of form: c dβ/dt
- Resulting expression: $I\ddot{\beta} + c\dot{\beta} + I\Omega^2\beta = A\sin\Omega t$
- Assuming the solution to be of form:

$$\beta = B\sin\Omega t + C\cos\Omega t$$

- Substituting and solving, we get

$$\beta = -\frac{A}{c\Omega}\cos\Omega t = \frac{A}{c\Omega}\sin\left(\Omega t - \pi/2\right)$$

Solving ODE (Partially)

- Solution $\beta = \frac{A}{c\Omega}\sin(\Omega t - {}^{\pi}/_{2})$ for forcing field $A\sin(\Omega t)$

➔response lags 90° from excitation force.

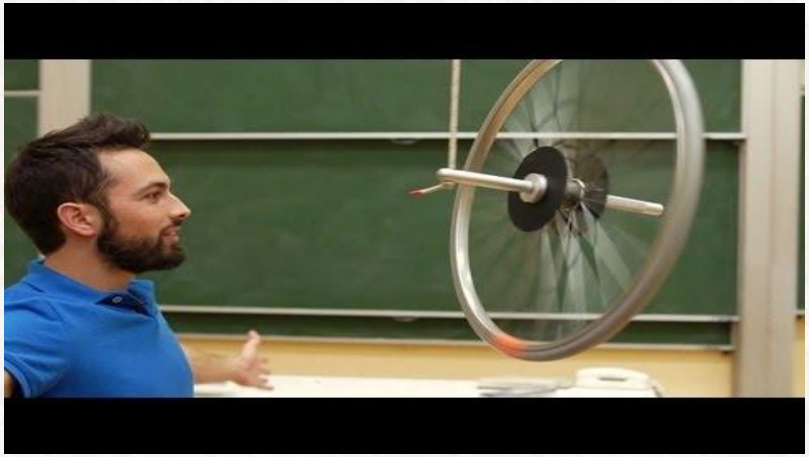
➔ the blades will flap upward to the highest level 90° after the highest pitch azimuth.

[Gyroscopic Precession!]

For example: In forward flight, the highest pitch setting is at $\Psi = 270^{o}$ and the lowest pitch setting is at $\Psi = 90^{o}$. Hence blades flap up at $\Psi = 0^{o}$ and down at $\Psi = 180^{o}$ (or in other words, TPP tilts forward). The overall thrust points forwards (to counter drag and to accelerate)

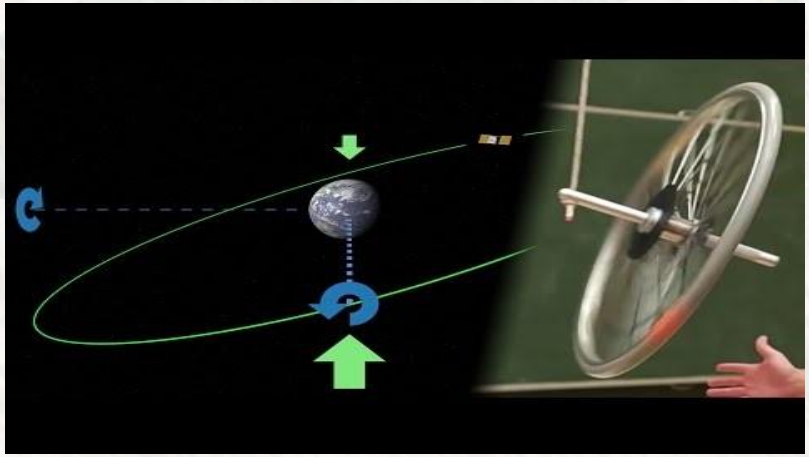
Gyroscopic Precession

- Please check out the following two videos on Gyroscopic Precession for a better understanding



Gyroscopic Precession

https://youtu.be/ty9QSiVC2g0



Gyroscopic precession -- An intuitive explanation

https://youtu.be/n5bKzBZ7XuM?t=204

General form of blade flapping motion

- The blade flapping dynamics ODE

$$I\ddot{\beta} + I\Omega^2\beta = \int_{r=0}^{r=R} \frac{1}{2}\rho c(\Omega r + V_\infty \sin\psi)^2 r C_l dr$$

- has solution of the form:

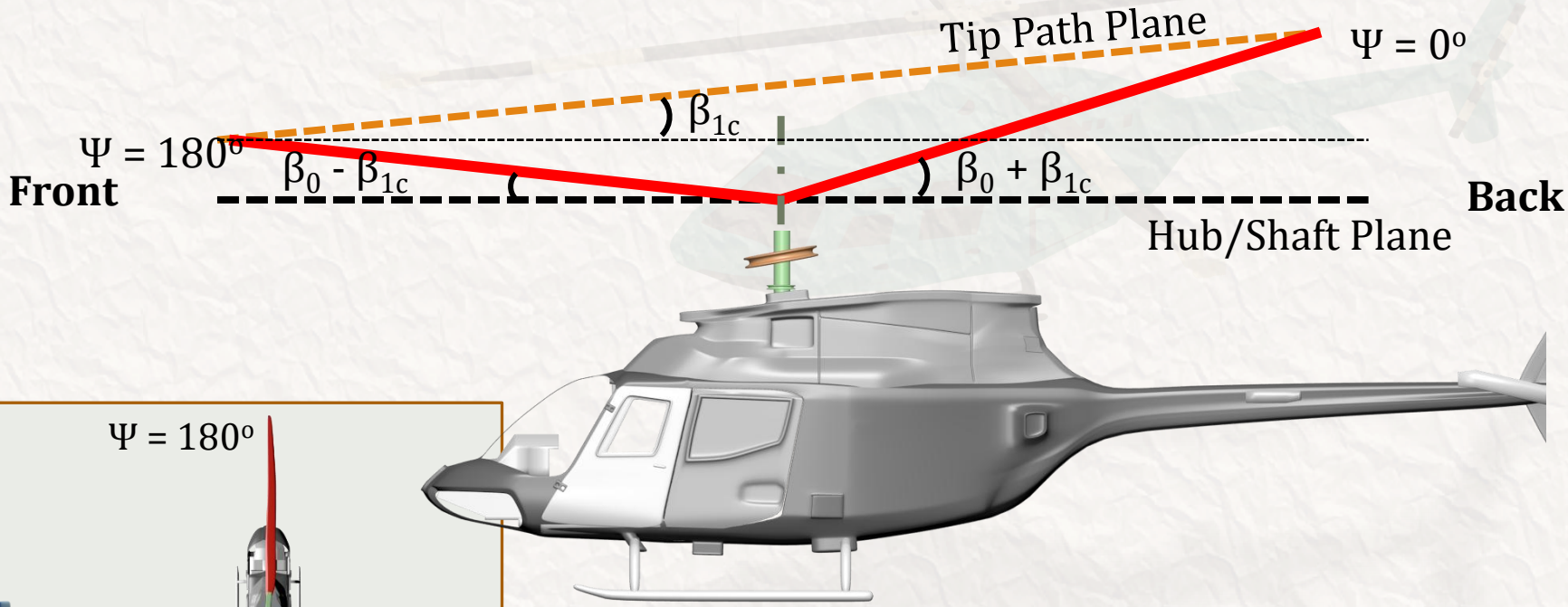
$$\beta = \beta_0 + \beta_{1c}\cos\psi + \beta_{1s}\sin\psi + \beta_{2c}\cos 2\psi + \beta_{2s}\sin 2\psi + \bullet\bullet\bullet$$

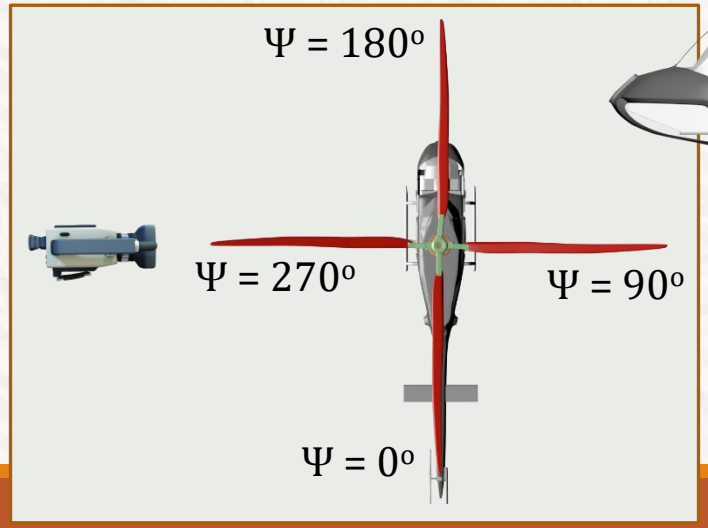
- For aerodynamics purpose, terms upto the first harmonic ψ are sufficient. Higher harmonics (2ψ and above) are important from dynamics and vibration perspective

Note that β_0 is the coning angle and $\psi = \Omega t$

Significance of β_{1c}

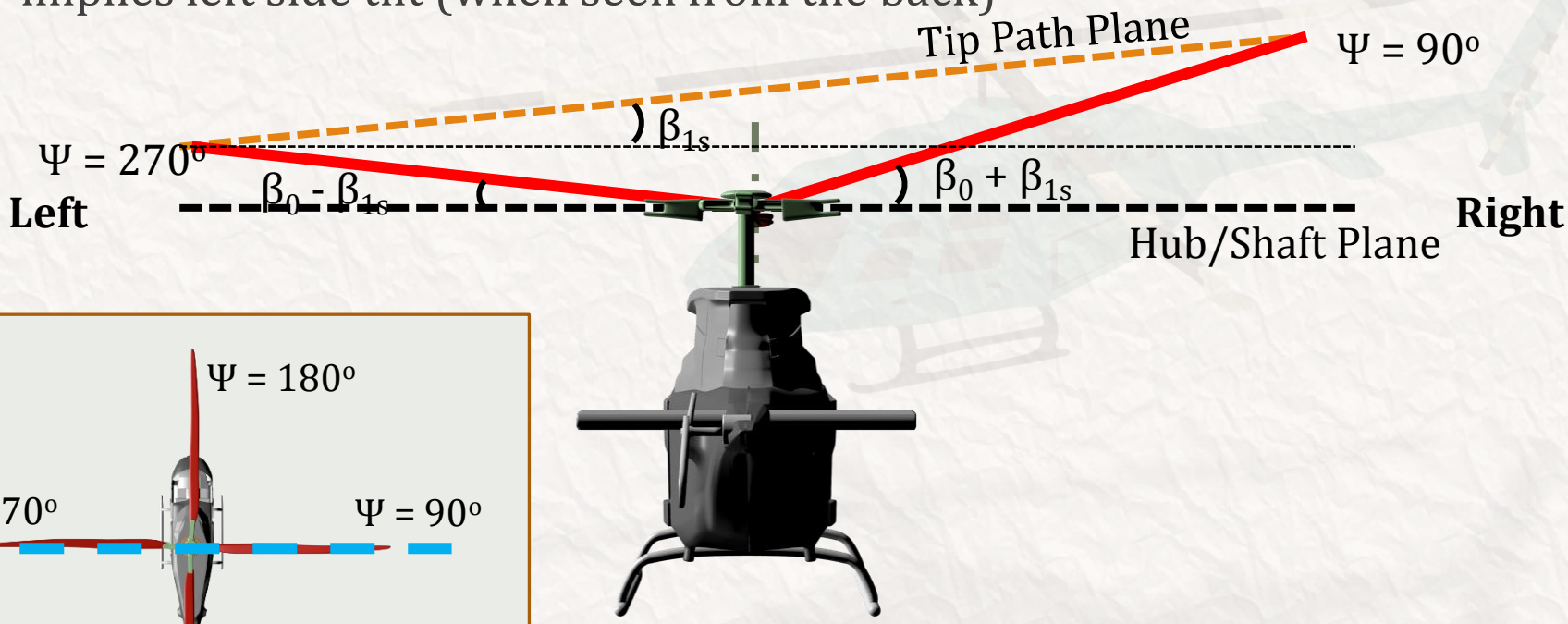
- **β_{1c}** is how much the TPP tilts forwards or backward w.r.t shaft plane. Positive β_{1c} implies forward tilt

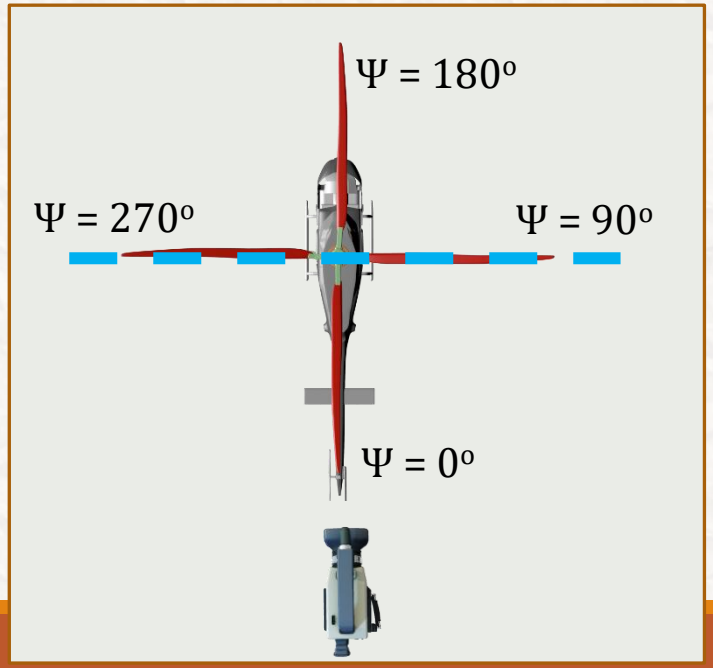




Significance of β_{1s}

- **β_{1s}** is how much the TPP tilts sidewards w.r.t shaft plane. Positive β_{1s} implies left side tilt (when seen from the back)





Pitch Control (Pilot Input)

- Pilot controls collective and cyclic pitches of the blades through swashplate.

$$\theta_{75\%R} = \theta_0 + \theta_{1c}\cos\psi + \theta_{1s}\sin\psi$$

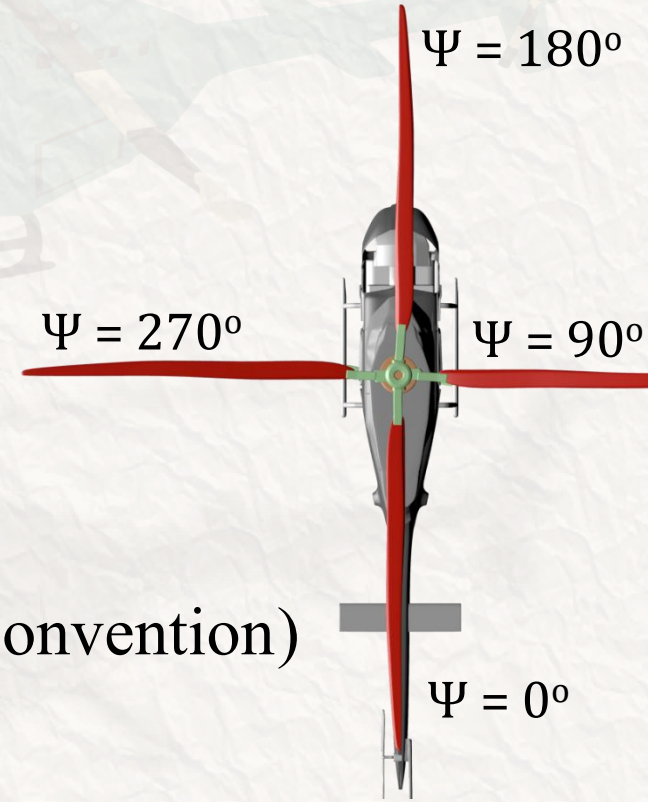
Where,

θ_0 = Collective Pitch

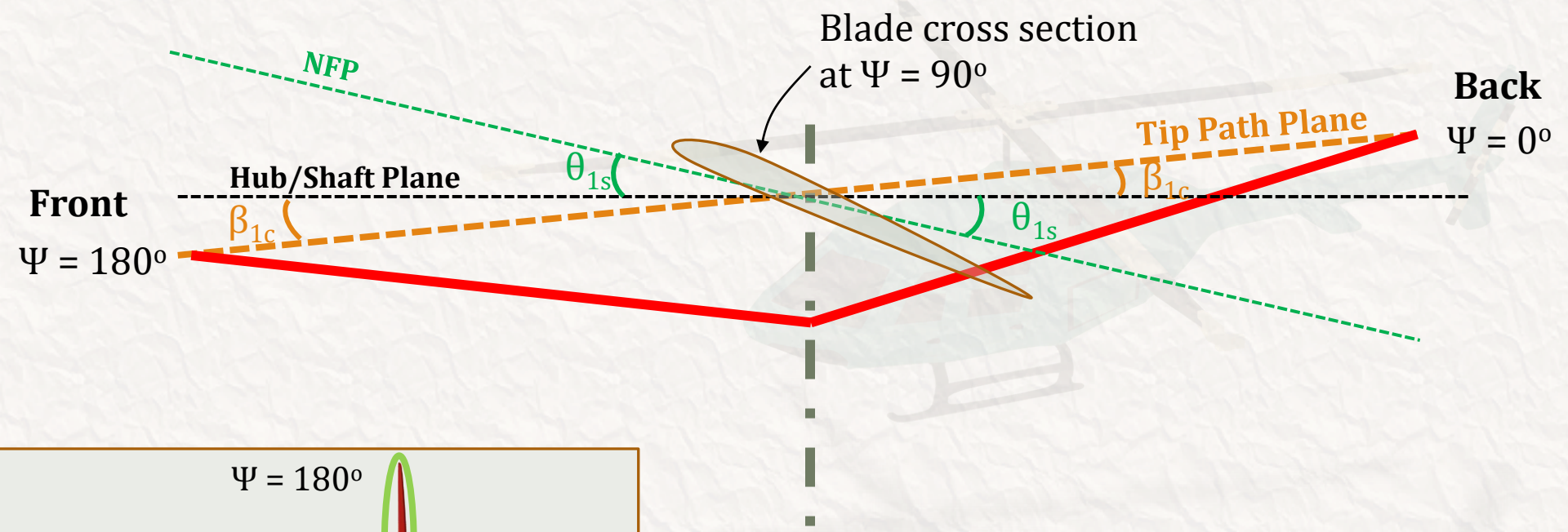
θ_{1c} = Lateral Cyclic (Side wards)

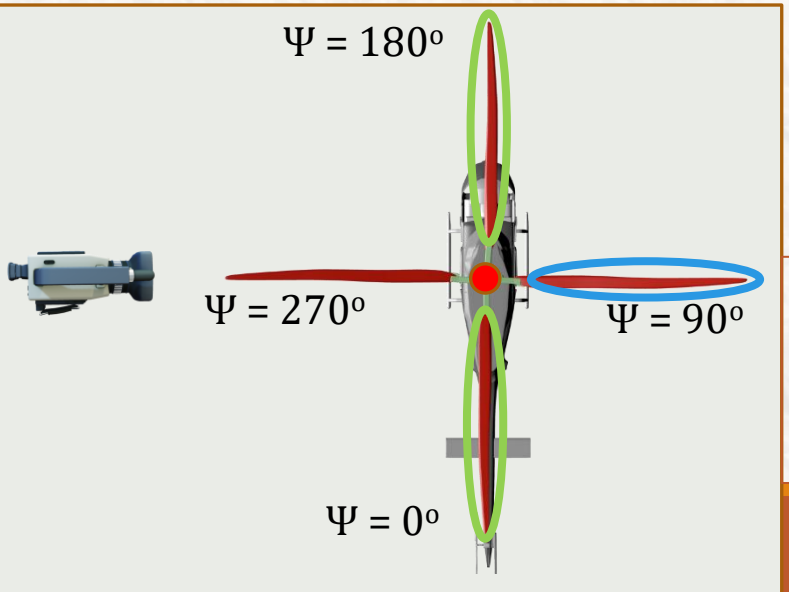
θ_{1s} = Longitudinal Cyclic (Fore & Aft)

$\theta_{75\%R}$ = Pitch at 75%R (definition as per convention)



Longitudinal Control and Angles

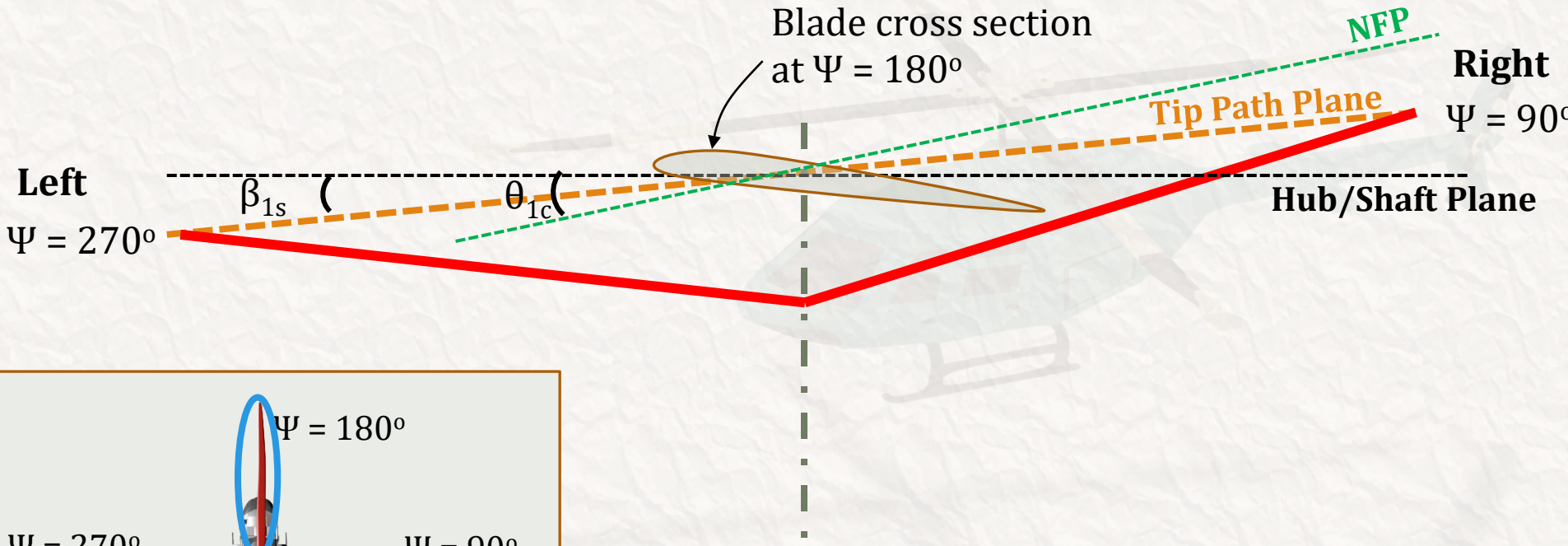


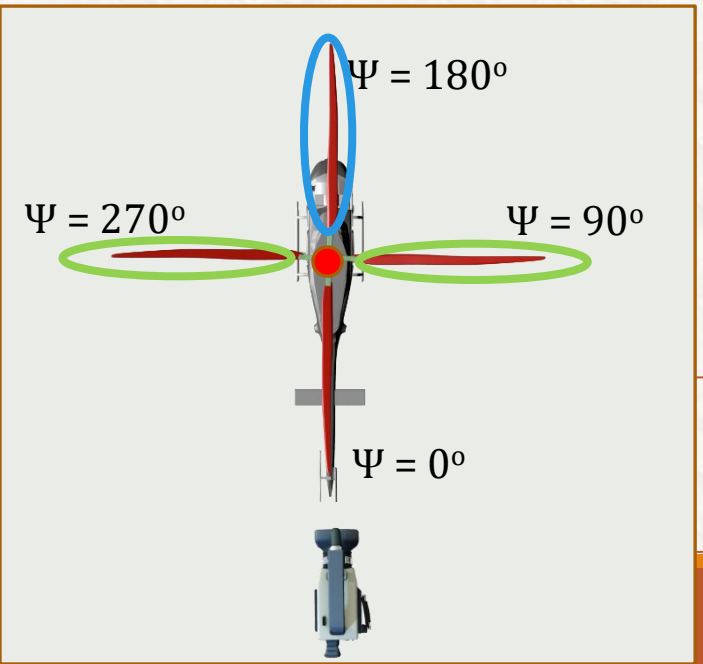


$$\theta = \theta_0 + \theta_{1c}\cos\psi + \theta_{1s}\sin\psi$$
$$\theta_{\psi=90^o} = \theta_0 + \theta_{1s}$$

$\theta_{1s} + \beta_{1c}$ is independent of coordinate system of measurement
(as the relative angle between the NFP and TPP does not change with coordinate system)

Lateral Control and Angles





$$\theta = \theta_0 + \theta_{1c}\cos\psi + \theta_{1s}\sin\psi$$
$$\theta_{\psi=180^o} = \theta_0 - \theta_{1c}$$

θ_{1c} - β_{1s} is independent of coordinate system of measurement (as the relative angle between the NFP and TPP does not change with coordinate system)

Angles in different coordinate systems

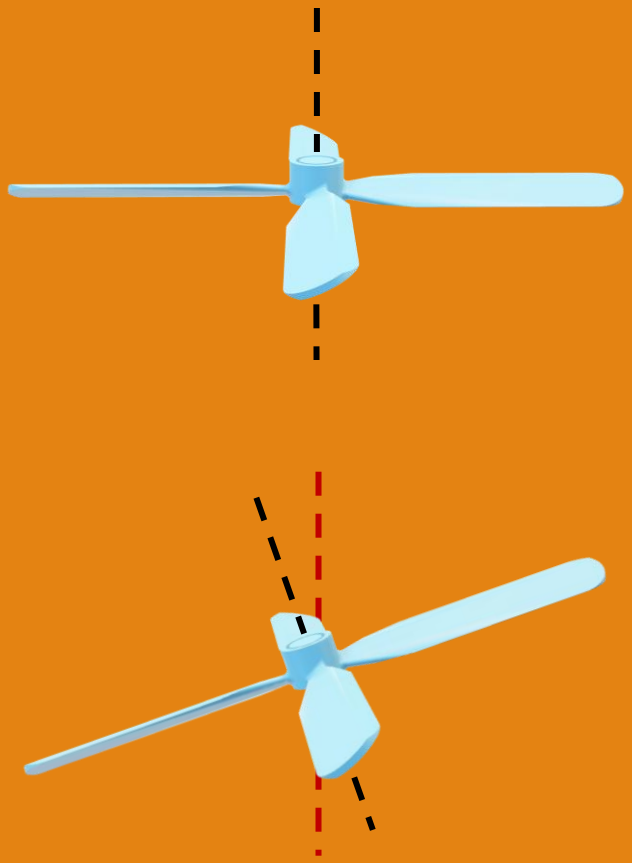
- $\theta|_{TPP} = \theta$ pitch as observed from Tip Path Plane
- $\beta|_{NFP} = \beta$ flap angle as obsered from No Feathering Plane

- $\beta_{1c} + \theta_{1s} = \beta_{1c}|_{NFP} = \theta_{1s}|_{TPP}$
- $\beta_{1s} - \theta_{1c} = \beta_{1s}|_{NFP} = -\theta_{1c}|_{TPP}$

- $\beta|_{TPP} = \beta_0$ and $\theta|_{NFP} = \theta_0$ by definition

$(\beta_{1c} + \theta_{1s})$ and $(\beta_{1s} - \theta_{1c})$ will always appear as single terms in the future derivations. That is so, because aerodynamics on a blade section is independent of whether the resultant pitch is due to direct pilot input (θ)or due to tilting of the TPP (β)

Equivalence of Flapping and Pitching Motions



All Planes Horizontal:

$\beta_0 = 5\ ;\ \beta_{1c} = 0$

$\theta_0 = 10\ ;\ \theta_{1s} = 0$

If entire shaft axis (black) is tilted by 5 degrees with frozen flapping and feathering:

$\beta_0 = 5\ ;\ \beta_{1c} = 0\ ;\quad \theta_0 = 10\ ;\ \theta_{1s} = 0$

Blade angle at 90^0 = 5; Blade angle at 270^0 = 15 (w.r.t horizontal)

If flapping + feathering enabled with frozen shaft axis (red):

$\beta_0 = 5\ ;\ \beta_{1c} = 5\ ;\quad \theta_0 = 10\ ;\ \theta_{1s} = -5$

Blade angle at 90^0 = 5; Blade angle at 270^0 = 15 (w.r.t horizontal)